

Neural Network Family - Complete Taxonomy

1. Basic Neural Networks

- Perceptron
- Adaline
- Multi-Layer Perceptron (MLP)
- Feedforward Neural Network (FNN)
- Deep Neural Network (DNN)

2. Convolutional Neural Networks (CNN Family)

Purpose: Computer Vision, Image Processing

- LeNet-5
- AlexNet
- ZFNet
- VGG16 / VGG19
- GoogLeNet (Inception)
- Inception-v2/v3/v4
- Xception
- ResNet
- ResNeXt
- Wide ResNet
- DenseNet
- SENet
- MobileNet
- ShuffleNet
- EfficientNet
- ConvNeXt
- U-Net
- Mask R-CNN Backbone Networks

3. Recurrent Neural Networks (RNN Family)

Purpose: Sequential Data, Time Series, NLP

- Simple RNN
- Elman Network
- Jordan Network
- Bidirectional RNN
- LSTM
- Bidirectional LSTM
- GRU
- Bidirectional GRU
- Seq2Seq Models
- Encoder-Decoder Networks

4. Attention Mechanisms

- Bahdanau Attention
- Luong Attention
- Self-Attention
- Multi-Head Attention

5. Transformer Family

Foundation Architecture:

- Transformer

Encoder Models:

- BERT

- RoBERTa
- DistilBERT
- ALBERT
- ELECTRA

Decoder Models:

- GPT-1
- GPT-2
- GPT-3
- GPT-4 Family
- LLaMA Family
- Mistral Family
- Falcon

Encoder-Decoder Models:

- T5
- BART
- Pegasus
- FLAN-T5

6. Autoencoder Family

- Basic Autoencoder
- Sparse Autoencoder
- Denoising Autoencoder
- Contractive Autoencoder
- Variational Autoencoder (VAE)
- Beta-VAE
- Vector Quantized VAE (VQ-VAE)

7. Generative Neural Networks

GAN Family:

- GAN
- DCGAN
- Conditional GAN
- CycleGAN
- Pix2Pix
- StyleGAN
- StyleGAN2
- StyleGAN3
- StarGAN

Diffusion Models:

- DDPM
- Stable Diffusion
- Latent Diffusion Models
- Imagen
- DALL-E Style Diffusion Systems

8. Graph Neural Networks (GNN)

- Graph Convolutional Network (GCN)
- Graph Attention Network (GAT)
- GraphSAGE
- Gated Graph Neural Network
- Message Passing Neural Network (MPNN)

9. Energy-Based / Probabilistic Neural Networks

- Restricted Boltzmann Machine (RBM)
- Deep Belief Network (DBN)
- Deep Boltzmann Machine (DBM)
- Hopfield Network

10. Specialized Neural Architectures

- Siamese Networks
- Triplet Networks
- Capsule Networks (CapsNet)
- Echo State Networks
- Reservoir Computing Networks
- Neural ODEs
- Memory Networks
- Pointer Networks
- Neural Turing Machines

11. Reinforcement Learning Neural Architectures

- Deep Q-Network (DQN)
- Double DQN
- Dueling DQN
- Deep Deterministic Policy Gradient (DDPG)
- PPO Networks
- Actor-Critic Networks
- A3C / A2C

Most Important Families for Exams & Industry

1. MLP / Feedforward Networks
2. CNNs
3. RNNs
4. LSTMs
5. GRUs
6. Autoencoders
7. GANs
8. Transformers
9. Graph Neural Networks
10. Diffusion Models